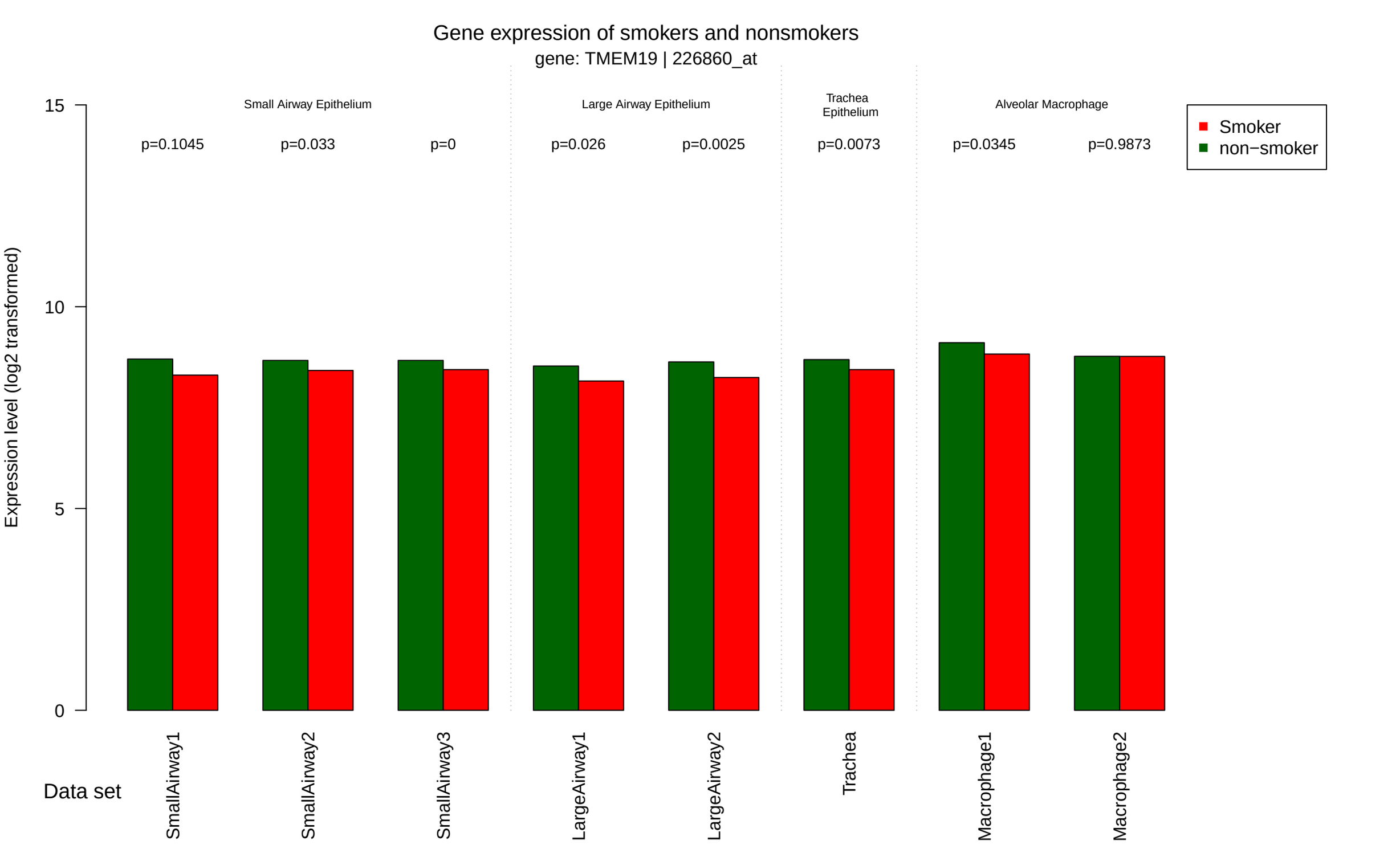
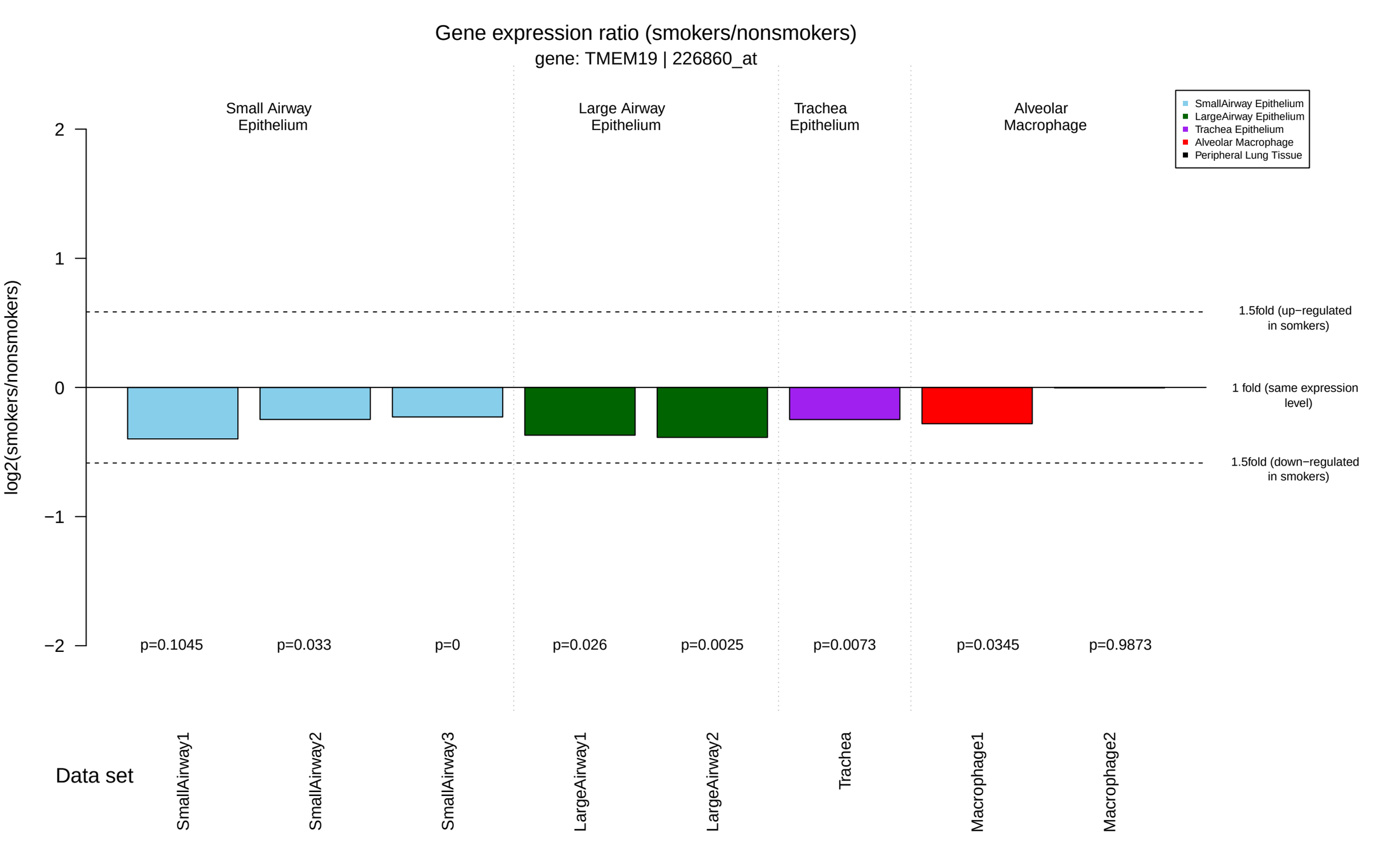
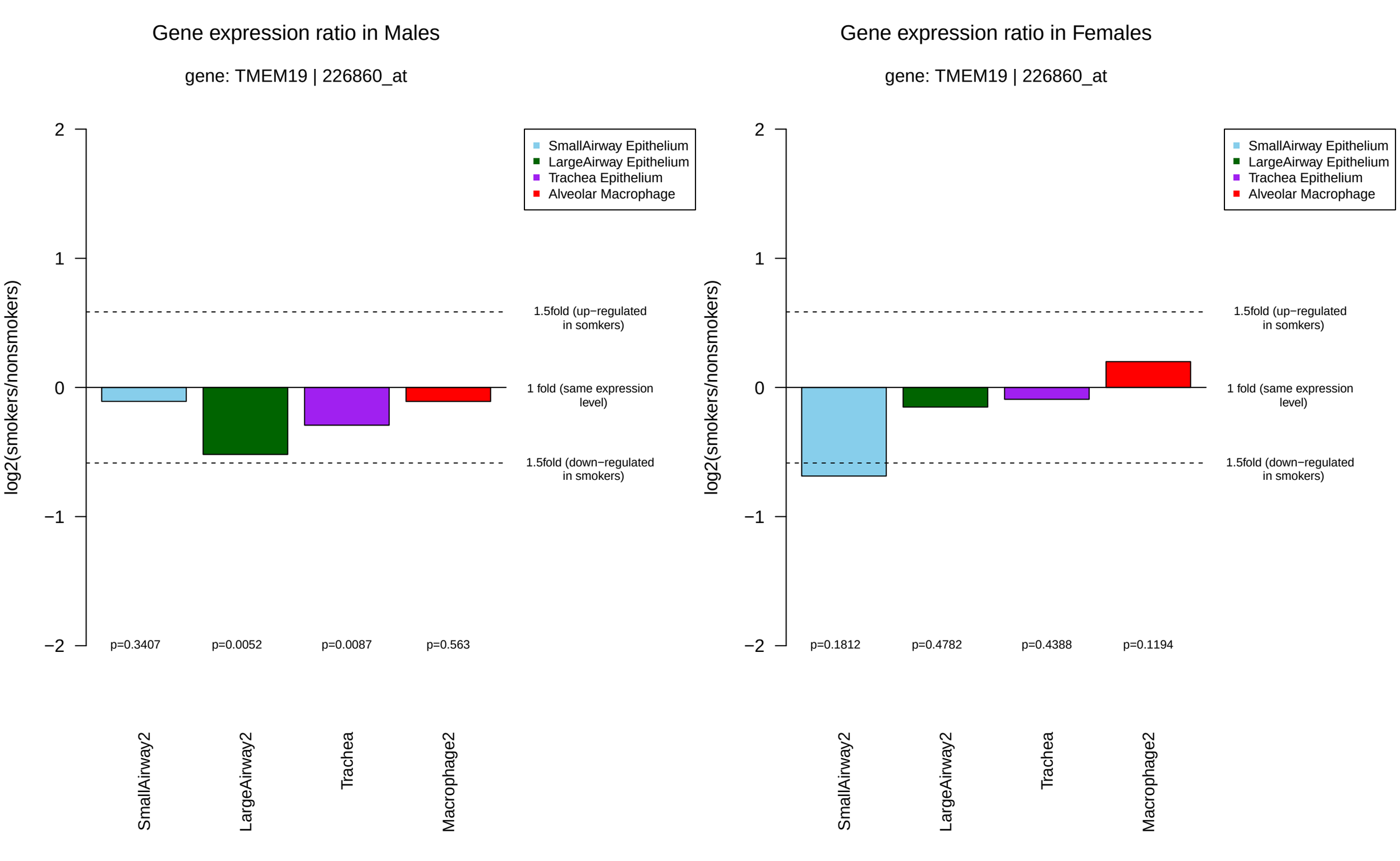
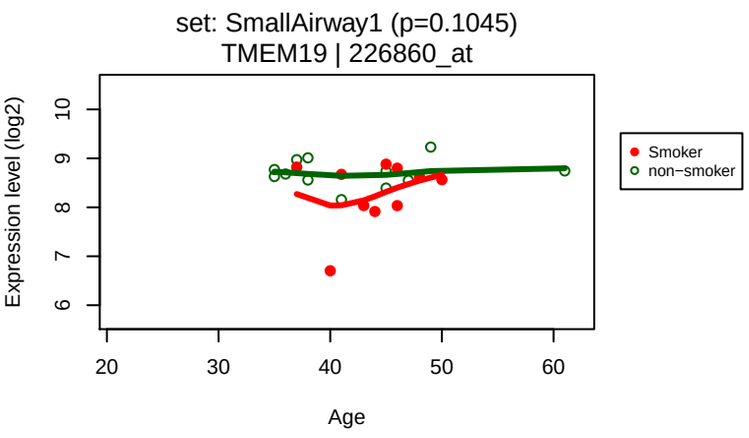
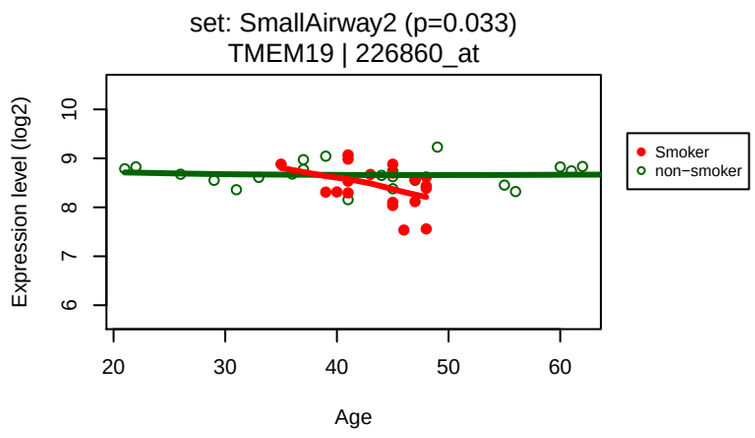
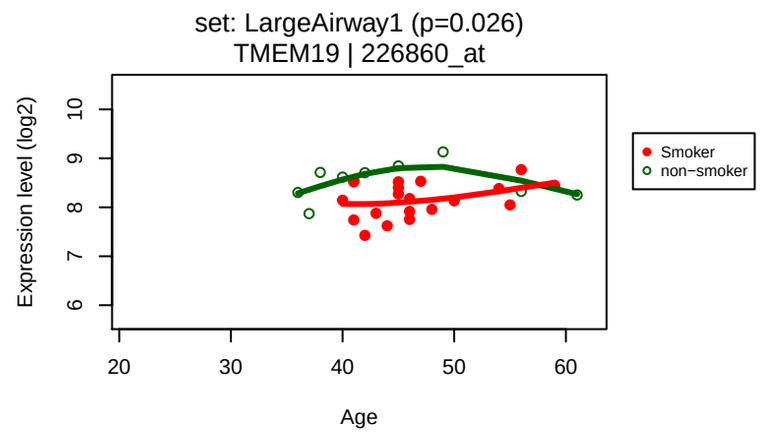

Gene expression of smokers and nonsmokers
gene: TMEM19 | 226860_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1045
p=0.033
p=0
p=0.026
p=0.0025
p=0.0073
p=0.0345
p=0.9873
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

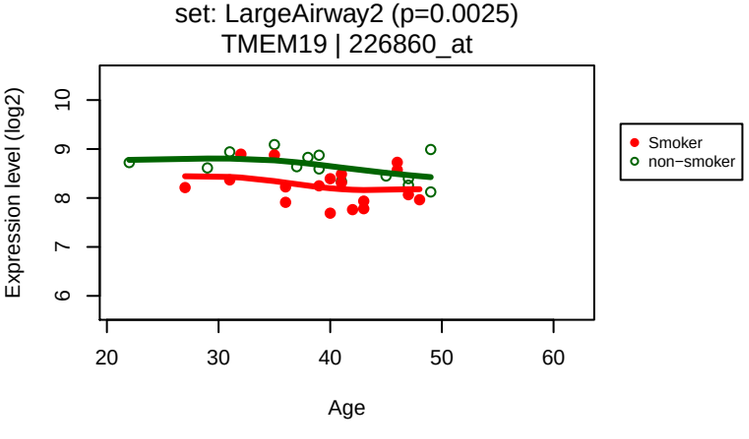

Gene expression ratio (smokers/nonsmokers)
gene: TMEM19 | 226860_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1045
p=0.033
p=0
p=0.026
p=0.0025
p=0.0073
p=0.0345
p=0.9873
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

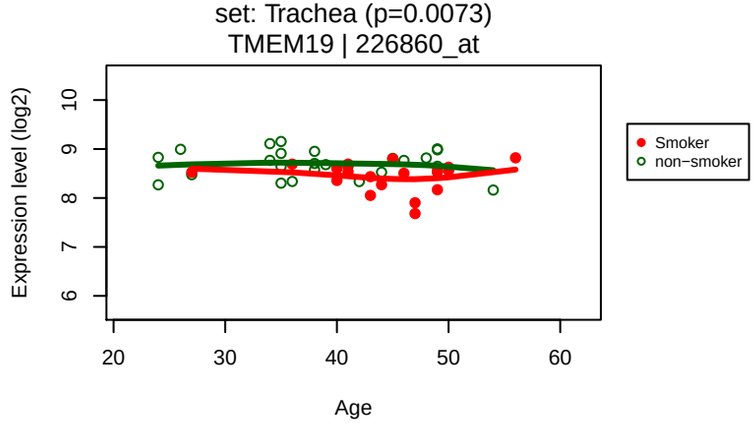

Gene expression ratio in Males
gene: TMEM19 | 226860_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3407
p=0.0052
p=0.0087
p=0.563
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM19 | 226860_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1812
p=0.4782
p=0.4388
p=0.1194
SmallAirway2
LargeAirway2
Trachea
Macrophage2

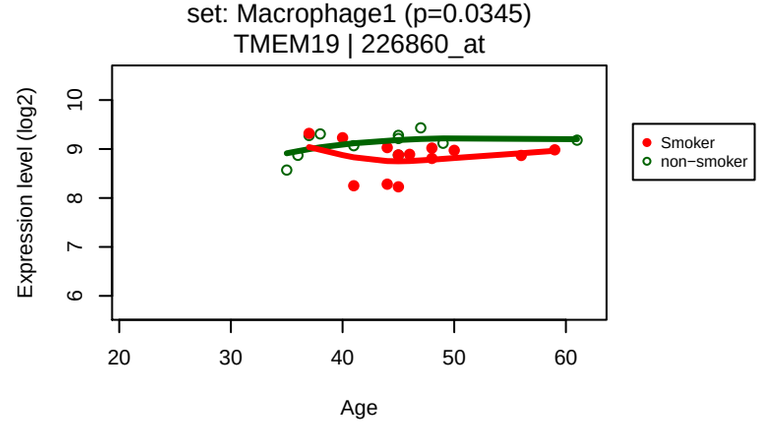

set: SmallAirway1 (p=0.1045)
TMEM19 | 226860_at
Expression level (log2)
Age
Smoker
non−smoker

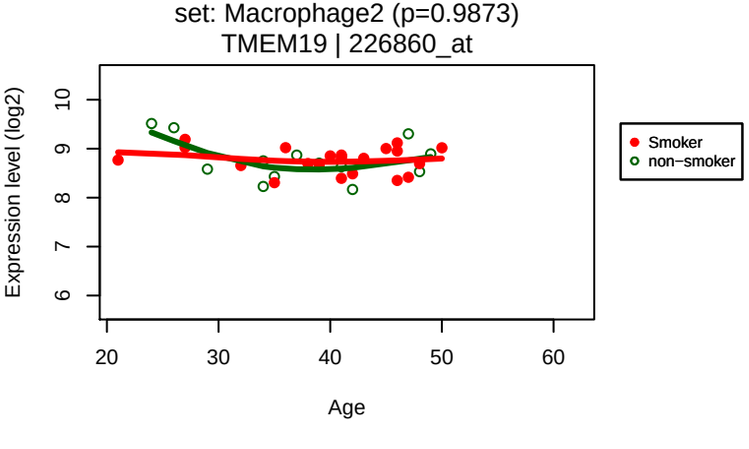

set: SmallAirway2 (p=0.033)
TMEM19 | 226860_at
Expression level (log2)
Age
Smoker
non−smoker

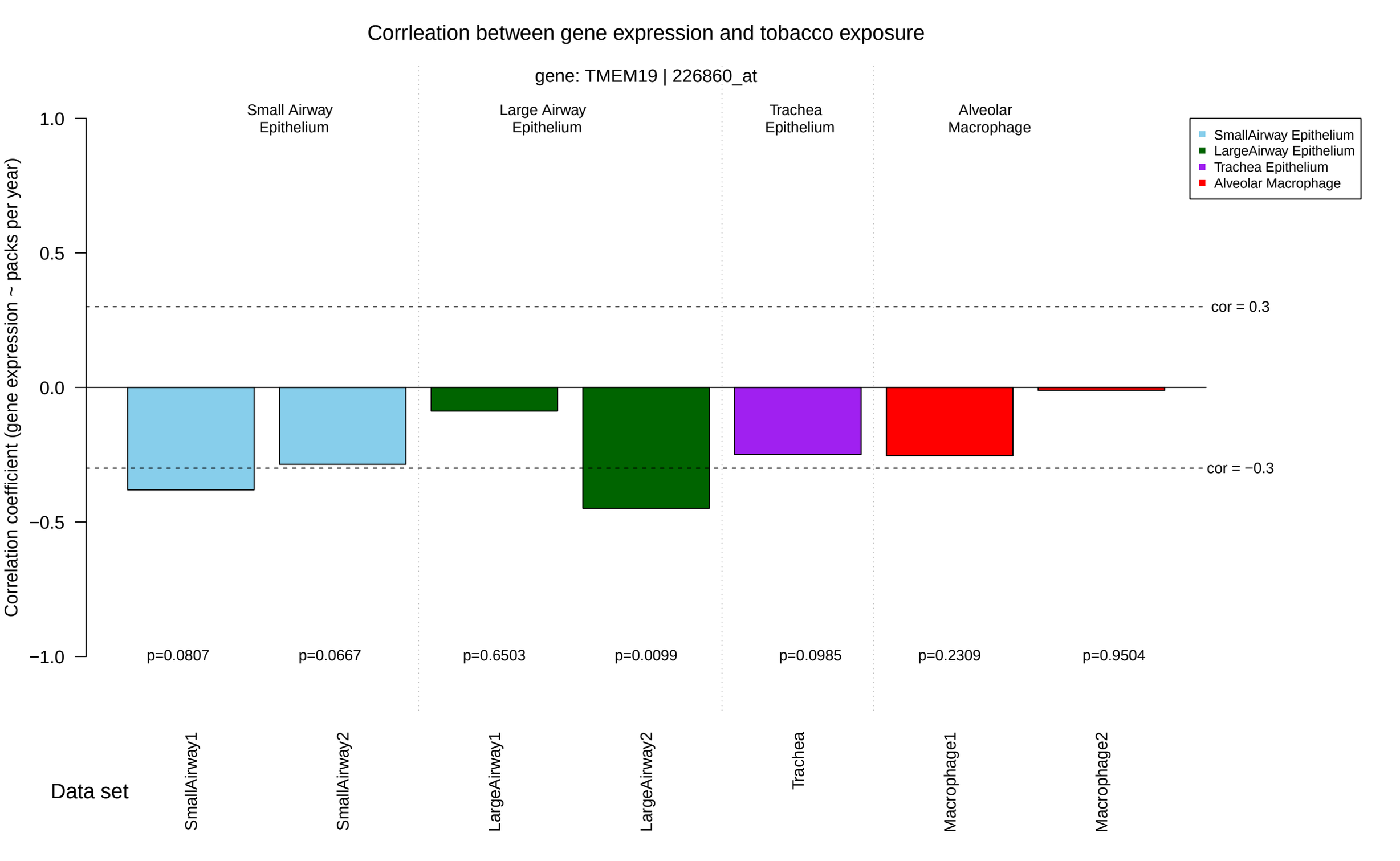

set: LargeAirway1 (p=0.026)
TMEM19 | 226860_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0025)
TMEM19 | 226860_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0073)
TMEM19 | 226860_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0345)
TMEM19 | 226860_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9873)
TMEM19 | 226860_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM19 | 226860_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0807
p=0.0667
p=0.6503
p=0.0099
p=0.0985
p=0.2309
p=0.9504
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2